Neural Computing and Applications (2025) 37:1–24
https://doi.org/10.1007/s00521-024-10453-2

ORIGINAL ARTICLE

Revolutionizing cardiovascular health: integrating deep learning techniques for predictive analysis of personal key indicators in heart disease

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Received: 14 February 2024 / Accepted: 9 September 2024 / Published online: 15 November 2024
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Abstract
Cardiovascular diseases (CVDs) remain a global burden, highlighting the need for innovative approaches for early detection and intervention. This study investigates the potential of deep learning, specifically convolutional neural networks (CNNs), to improve the prediction of heart disease risk using key personal health markers. Our approach revolutionizes traditional healthcare predictive modeling by integrating CNNs, which excel at uncovering subtle patterns and hidden interactions among various health indicators such as blood pressure, cholesterol levels, and lifestyle factors. To achieve this, we leverage advanced neural network architectures. The model utilizes embedding layers to transform categorical data into numerical representations, convolutional layers to extract spatial features, and dense layers to model complex interactions and predict CVD risk. Regularization techniques like dropout and batch normalization, along with hyperparameter optimization, enhance model generalizability and performance. Rigorous validation against conventional methods demonstrates the model's superiority, with a significantly higher R^2 value of 0.994. This achievement underscores the model's potential as a valuable tool for clinicians in CVD prevention and management. The study also emphasizes the need for interpretability in deep learning models and addresses ethical considerations to ensure responsible implementation in clinical practice.

Keyword Cardiovascular diseases; Deep learning; Predictive analysis; Health indicators

1 Introduction

Cardiovascular diseases (CVDs), encompassing conditions like heart attack and stroke, are the world's leading killer. In 2019, an estimated 17.9 million deaths, or 32% of all global fatalities, were attributed to CVDs. Notably, 85% of these deaths were caused by heart attack or stroke as illustrated in Fig. 1.

The burden of CVDs is particularly heavy in low- and middle-income countries, where over three-quarters of CVD deaths occur. Additionally, in 2019, CVDs were responsible for 38% of the 17 million premature deaths (before age 70) from non-communicable diseases [1].

The good news is that most cardiovascular diseases are preventable. By addressing lifestyle risk factors such as tobacco use, unhealthy diet, physical inactivity, excessive weight, and harmful alcohol consumption, we can significantly reduce the prevalence of CVDs. Early detection is also crucial, allowing for timely intervention with counseling and medication to manage the disease.

CVDs cast a long shadow over global health, claiming millions of lives annually and imposing a significant burden on healthcare systems worldwide [1]. Early detection and intervention are paramount in combating CVDs, allowing for timely treatment and improved patient outcomes [2]. Fortunately, advancements in medical research offer promising avenues for achieving this goal. Deep learning algorithms have emerged as a transformative force in healthcare analytics, demonstrating exceptional capabilities in predictive modeling and pattern recognition [3].

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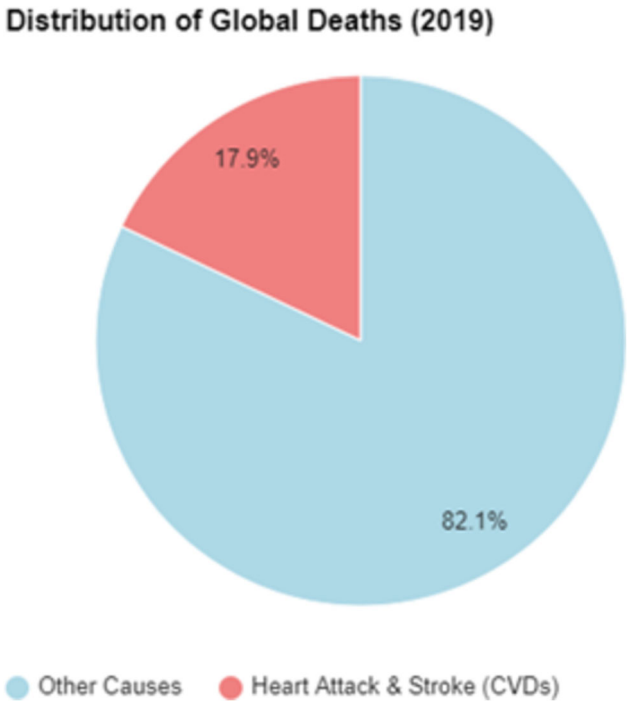


Fig. 1 Global Distribution of CVDs in 2019

Their remarkable success in fields like computer vision and natural language processing fuels excitement for their potential application in health care, particularly in predicting the onset of cardiovascular diseases [4]. Several risk factors for CVDs are identified in Table 1, including tobacco use, unhealthy diet, and physical inactivity.

Beyond the sheer number of lives impacted, CVDs inflict a devastating toll on individuals and families. The emotional and physical strain of living with a chronic heart condition can be immense. Early detection empowers individuals to take a proactive role in managing their health, potentially preventing the disease's progression and improving their quality of life [5]. This highlights the critical need for more accurate and personalized risk assessment tools.

However, traditional CVD risk assessment methods often rely on a limited set of parameters, potentially overlooking the complex interplay between various lifestyle factors and their cumulative impact on heart health. Factors like physical activity levels, dietary habits, smoking history, and even sleep patterns can all contribute to an individual's risk profile [6, 7]. While existing research has established correlations between these factors and cardiovascular outcomes, a deeper understanding of how they interact at the individual level remains elusive. This gap in knowledge hinders the development of truly personalized interventions tailored to each patient's unique risk profile [8, 9]. As highlighted in Table 2, these limitations of conventional methods necessitate a more holistic approach to CVD risk assessment.

To address these shortcomings, this study delves into the potential of deep learning, specifically convolutional neural networks (CNNs), to revolutionize CVD risk assessment. CNNs excel at extracting complex patterns from diverse data formats, making them ideally suited for analyzing the rich tapestry of personal health information [10]. By leveraging the power of CNNs, we aim to develop a novel methodology that can uncover subtle interactions between various lifestyle choices, dietary habits, and personal health markers, ultimately leading to a more comprehensive understanding of individual CVD risk.

1.1 Problem statement

Current predictive models in cardiovascular health often fail to capture the unique nature of risk factors, overlooking the intricate interplay between personal health metrics. This results in inaccuracies when assessing individual CVD risk. Additionally, the potential of deep learning techniques, particularly CNNs, to analyze these complex interactions remains underexplored.

1.2 Main contributions

This study presents a novel approach that integrates deep learning with a curated dataset of personal health information. We aim to achieve the following key contributions:

- **Novel Integration of Deep Learning:** We propose a novel integration of deep learning approaches (CNNs)

Table 1 Risk factors for cardiovascular diseases (CVDs)

Risk factor	Description
Age	Increasing age is a significant risk factor for CVDs
Gender	Men are generally at higher risk for CVDs compared to women
Family history	Having a family history of CVDs increases an individual's risk
Lifestyle factors	Smoking, unhealthy diet, physical inactivity, and excessive alcohol consumption all contribute to CVD risk
Medical conditions	High blood pressure, high cholesterol, diabetes, and obesity are major risk factors

Table 2 Limitations of Conventional CVD Risk Assessment Methods

Limitation	Description
Limited parameters	Existing methods often focus on a restricted set of risk factors, neglecting complex interactions
Inaccurate risk stratification	Inability to capture the cumulative impact of various lifestyle factors on individual risk profiles
Difficulty in modeling non-linear relationships	Statistical models may struggle to identify intricate non-linear relationships between health indicators

into the traditional predictive modeling framework for CVD risk assessment.

- **Enhanced Predictive Accuracy:** We aim to utilize CNNs and a curated dataset to improve the accuracy of identifying individuals at risk of heart disease.
- **Uncovering Subtle Interactions:** This study goes beyond conventional methods by aiming to reveal subtle interactions between personal health metrics, providing a deeper understanding of how lifestyle and dietary choices influence heart health.
- **Potential for Revolutionizing Preventive Care:** Our findings have the potential to significantly transform preventive health care for CVDs by enabling more precise and individualized risk assessments, facilitating proactive interventions, and informing personalized treatment plans.
- **Addressing Ethical Considerations:** We acknowledge the ethical concerns surrounding deep learning applications in health care and ensure transparency, interpretability, and equity in model development and use.

In summary, CVDs are the leading global cause of death, significantly impacting low- and middle-income countries. Traditional risk assessment methods often fail to capture the complex interactions between various lifestyle factors and health metrics. This study leverages CNNs to improve predictive accuracy and uncover subtle interactions in personal health data, aiming to enhance individualized CVD risk assessments and promote more effective preventive healthcare strategies.

The remainder of this paper is organized as follows: Sect. 2 presents the literature review, Sect. 3 presents the main contribution, Sect. 4 discusses the proposed method, Sect. 5 details the experimental evaluation, and Sect. 6 concludes the work.

2 Literature review

Heart disease is one of the leading causes of death worldwide. The use of artificial intelligence (AI) techniques allows for the early detection of cardiac disease by monitoring specific parameters such as blood pressure, body weight, cholesterol, sugar level, and heart rate [10]. While ML and DL algorithms are transforming the present healthcare system, effectively and reliably predicting heart illness remains a challenge. A variety of categorization approaches have been used to predict heart disease. The ensemble learning technique, specifically random forest (RF), has demonstrated promising results in predicting cardiac disease [11].

The study [12] used support vector machines (SVM) for classification after applying feature selection approaches such as the Fisher score and Matthew's correlation. In a study conducted in [13], a deep learning system named DeepLabeler was developed to automatically label ICD-9 codes. Their proposed system captures and encodes both local and global data using the document-to-vector (D2V) approach and a CNN. The model's two main features are multi-label classification and feature extraction. The study conducted in [14] used the reverse time attention model (RETAIN), which integrates an attention mechanism and is based on a combination of recurrent neural networks (RNNs).

This enables the model to focus on the most important qualities or time periods in the input sequence. By assigning a weighted relevance value to each attribute or time step in the sequence, we can better comprehend RETAIN. In this method, clinicians and specialists can determine which components or time sequences are most important for the model's predictions. Current cutting-edge deep learning models lack great feature extraction capabilities in complex and noisy environments, limiting the development of exact and consistent object differentiation [15]. Previous research can be largely classified into two categories: deep learning approaches and standard shallow approaches [16].

A robust and high-performing deep learning (DL) model for the exact identification of valve heart disorders (VHDs) was provided [17]. The study published in [18] created a model for predicting the likelihood of CVD in their population using data from a Japanese urban cohort study. The system for diagnosing coronary disease and stroke was built with multivariable Cox proportional hazard approaches. Using their proposed technique, they were able to

investigate a range of parameters and create a credible model for assessing the risk of CVD events.

A novel ML approach for predicting heart disease was developed in the study described in [19]. They used the Cleveland heart disease dataset to test RF and decision tree (DT) techniques. Their testing results revealed an accuracy of 88.7% in detecting cardiac disease. Numerous machine learning approaches were used to examine huge and complex medical data, supporting healthcare workers in the early detection of heart disease [20]. The study applied a range of classification models, including DT, naive Bayes (NB), K-nearest neighbor (KNN), and RF algorithm, to compute a variety of heart disease-related problems. Their primary goal was to evaluate the likelihood of people having a heart attack in the future.

The study, conducted in 2013, tried to diagnose heart failure using sequential electronic health record (EHR) data. They used real-world datasets that included data from hospital departments, health records, and patient diagnostic information related to cardiac disorders. The primary goal of their research was to accurately detect and classify persons at risk of heart failure using full EHR data. The effectiveness of combining tree-based ensemble methods and the synthetic minority oversampling technique (SMOTE) was investigated [21]. This method was developed to address the issue of data imbalance in heart failure patient survival prediction.

The study's goal is to improve the accuracy of forecasting survival outcomes for patients with heart failure by employing ensemble methods and SMOTE to rebalance the data. The study published in [22] used a hybrid model that included clustering and classification to predict type 2 diabetes. The initial part of this model's two-step process was K-means clustering, which was followed by the C4.5 classification strategy, which employed a k-fold cross-validation method. The proposed hybrid technique yielded promising results, with a classification rate of 88.38%. This concept has huge potential for clinicians since it allows them to make well-informed clinical decisions concerning diabetes care.

The most recent research shows the numerous strategies utilized to improve heart disease prediction accuracy. Researchers have made significant progress in enhancing the precision and effectiveness of prediction models by employing ensemble learning [23], feature extraction [24], DL models [25], and other methods. To address the shortcomings of previous research, a novel strategy for predicting cardiac disease is proposed, which employs a self-attention-based transformer model. This cutting-edge model was specifically designed to address the difficulties associated with investigating and forecasting heart illness. The program accurately captures complex patterns and relationships in medical data using self-attention processes, allowing for more exact predictions. Table 3 presents a comparison of various algorithms.

Table 3 gives a comprehensive review of numerous algorithms used to predict cardiac disease, including algorithmic explanations, benefits, and downsides. CardioNet-Deep, which uses CNNs, stands out for its capacity to discover subtle patterns and correlations in personal key markers associated with heart disease. However, its reliance on large and diverse datasets may present practical issues. The proposed Self-Attention Model takes a novel approach with the self-attention-based transformer model, exhibiting accurate capture of complex patterns and correlations in medical data. Nonetheless, the computational energy required to develop and train transformer models is recognized as a possible disadvantage.

CNNs are effective in capturing spatial patterns, particularly in medical imaging, but they may require large computer resources. RNNs are suitable for sequential health data but may have problems with vanishing/exploding gradients. LSTM networks are effective at handling sequence data and long-term dependencies, but they encounter computational complexity difficulties. This comparison analysis is useful for academics and practitioners who want to choose the best algorithm for their individual needs and restrictions in the domain of cardiovascular health prediction.

The research gaps in previous algorithms can be summarized in the following:

- Limited investigation on the complex interplay between lifestyle choices, food habits, and customized heart health measures.
- There is little study on the subtle interactions between numerous personal vital indicators, making it difficult to gain a complete picture.
- Lack of sophisticated approaches for spotting small changes within individual health measures. Insufficient research into the specific interactions between deep learning models and personal key indicators in the context of cardiovascular health.
- There is a lack of emphasis on interpretability and ethical considerations when integrating advanced AI techniques, such as CNNs, in health care for heart disease prediction.
- Individuals have a limited awareness of the intricate relationships that exist between lifestyle, dietary choices, and cardiovascular health.
- There is a lack of research that bridges the gap between broad links revealed in lifestyle and cardiovascular risk and unique details at the individual level.
- More research using advanced AI methodologies, such as CNNs, is required to unravel complex linkages and

Table 3 Comparison of ML and DL techniques for cardiovascular disease prediction

Model	Algorithm description	Pros	Cons
CardioNet-Deep [26]	A method that uses deep learning techniques, notably convolutional neural networks (CNNs), to predict personal critical markers in heart disease	Detects subtle patterns and connections between health-related data	To ensure optimal training, big and diverse datasets are required
Proposed self-attention transformer model [27]	A unique technique to predicting cardiac disease that uses a self-attention-based transformer model that is intended to solve the complexity of investigating and forecasting heart disease	Accurately captures complicated patterns and correlations in medical data	The implementation and training of transformer models can be computationally intensive
Convolutional neural networks (CNNs) [28]	A deep learning model intended for image and sequence data extracts features using convolutional layers	Effective at capturing spatial patterns, and commonly utilized in medical imaging	May necessitate significant computational resources
Recurrent Neural Networks (RNNs) [29]	Deep learning model for sequence data that captures temporal dependencies	Suitable for sequential health data; captures time-varying patterns	There may be issues with vanishing/exploding gradients
Long Short-Term Memory (LSTM) Networks [30]	A form of RNN with memory cells intended to solve the vanishing gradient problem	Effective for sequence data; captures long-term dependencies	Computational complexity

provide a full understanding of how lifestyle and dietary factors affect heart health.

In-depth Exploration of Lifestyle Dynamics Using its self-attention-based transformer model, the proposed algorithm fills a research gap by exploring lifestyle choices and eating patterns in greater depth. This approach is specifically developed to examine and comprehend the complex relationships between various lifestyle factors and individual health measurements.

Detection of Subtle patterns By adding self-attention processes, the system excels at detecting subtle patterns in personal health measurements. Unlike classic models, the self-attention mechanism provides for a more detailed examination of individual indicators, allowing for the detection of complex patterns that may lead to cardiovascular health risks.

Sophisticated Detection Techniques The algorithm includes a sophisticated technique for detecting tiny patterns in personal health metrics. This fills a gap by offering a more sophisticated approach to understanding the links between lifestyle choices, food habits, and heart health at the individual level.

Comprehensive Understanding of Interaction Using the power of self-attention processes, the algorithm adds to a better understanding of how lifestyle and dietary factors affect heart health. It extends beyond broad linkages to unravel nuanced links at the individual level, bridging the knowledge gap about the customized nature of these interactions.

2.1 Existing methods and their limitations

Various approaches have been explored for predicting cardiovascular diseases (CVDs), each with distinct strengths and limitations:

- **Ensemble Learning:** Techniques such as random forests have shown promise in heart disease identification [11]. However, they may struggle with capturing intricate relationships between diverse health data points.
- **Support vector machines (SVMs):** Effective in classification tasks, SVMs require meticulous feature selection and may lack interpretability [12].
- **Deep learning models:** Models like DeepLabeler excel in feature extraction and multi-label classification [13]. However, their "black box" nature poses challenges for interpretability and trust in healthcare applications.
- **Recurrent neural networks (RNNs) and long short-term memory (LSTM) Networks:** These models are adept at handling sequential data but can be computationally intensive and susceptible to gradient problems [14].

2.2 Recent advancements and their implications

The field of cardiovascular health prediction continues to evolve with recent advancements:

- **Attention Mechanisms:** Integrated within deep learning models, attention mechanisms enhance

interpretability by focusing on critical features in the data [15].
- **Transformer Models:** Known for capturing long-range dependencies, transformer models offer potential in modeling complex interactions within health data [16].

2.3 Research gaps and proposed approach

Despite these advancements, significant gaps remain:

- **Limited Understanding of Complex Interactions:** Current research often focuses on isolated risk factors or general lifestyle associations [17, 18]. Understanding individual level interactions between lifestyle choices, dietary habits, and health metrics is essential for personalized risk assessment.
- **Need for Interpretability and Explainability:** Concerns persist regarding the opacity of AI models in healthcare. Techniques ensuring transparency in model predictions are crucial for ethical and trustworthy applications [19].

Our proposed approach addresses these gaps through a self-attention-based transformer model:

- **In-depth Exploration of Lifestyle Dynamics:** Leveraging self-attention, our model explores interactions between lifestyle factors and health metrics, revealing nuanced influences on CVD risk.
- **Detection of Subtle Patterns:** By identifying intricate patterns in personal health data, our model enhances sensitivity to subtle changes.
- **Comprehensive Understanding of Interactions:** Analyzing self-attention weights provides insights into how individual lifestyle choices contribute to CVD risk, promoting a deeper understanding of health dynamics.

3 CardioSentiNet: cardiovascular sentiment network

CardioSentiNet is an innovative deep learning technology that uses personal key factors to forecast heart disease. CardioSentiNet, which uses powerful self-attention-based transformer models, excels at capturing subtle patterns and correlations within individual health measurements. The algorithm promotes interpretability, making predictions that are clear and understandable to healthcare experts and end users. CardioSentiNet converts raw healthcare data into useful insights by doing extensive data pretreatment, feature engineering, and optimization. Its self-attention technique enables it to focus on crucial signs, allowing for accurate risk assessment for cardiovascular illnesses.

3.1 Rationale for utilizing convolutional neural networks (CNNs)

Traditionally, deep learning models for cardiovascular disease (CVD) risk prediction have primarily relied on fully-connected neural networks (FCNNs). While FCNNs have demonstrated success in various tasks, they may struggle to capture the inherent spatial relationships within complex data structures like personal health information. For example, analyzing sequential data points in an electrocardiogram (ECG) recording or identifying spatial patterns in medical images requires an understanding of the order and relative positions of these data points.

In contrast, CNNs are specifically designed to excel in extracting features from such data structures. CNNs utilize convolutional layers that employ filters to identify local patterns within the data. By applying these filters across the entire data input, they can effectively capture spatial relationships between data points, offering a significant advantage over FCNNs for tasks involving sequential or image-based data.

3.2 Novelty of the proposed approach

Our study proposes a novel and innovative approach to CVD risk assessment by integrating CNNs into the predictive modeling framework. This approach offers several key advantages over existing methods:

- **Enhanced Capacity for Feature Extraction:** By leveraging convolutional layers, our CNN architecture can effectively capture intricate spatial relationships within personal health data. This allows the model to identify subtle patterns and dependencies that might be missed by FCNNs, potentially leading to more accurate CVD risk predictions.
- **Improved Interpretability:** Our approach incorporates techniques to enhance the interpretability of the CNN model. This allows healthcare professionals to understand the rationale behind the model's predictions, fostering trust and facilitating informed decision-making.

By integrating CNNs and employing interpretability techniques, our proposed approach offers a distinct and valuable contribution to the field of CVD risk prediction. The model's ability to capture complex relationships within health data, coupled with its interpretability, empowers healthcare professionals with a powerful tool for identifying individuals at risk of heart disease.

CardioSentiNet, with a focus on ethics, privacy, and security, intends to transform cardiovascular health by offering accurate, tailored, and interpretable forecasts, thereby contributing to proactive healthcare and worldwide disease prevention. CardioSentiNet as depicted in Fig. 2 consists of six main phases; (i) data preprocessing and feature engineering, (ii) self-attention-based transformation, (iii) training and optimization, (iv) interpretability and ethical considerations, (v) validation and comparative analysis, (vi) result interpretation and model explainability.

Figure 3 illustrates the workflow of our proposed CardioSentiNet model for CVD risk prediction. The process begins with data preprocessing, where the raw data undergo cleaning, handling missing values, and necessary transformations. The preprocessed data then enter the self-attention-based transformation module, where self-attention mechanisms capture complex relationships within the data. These enriched data feed into the CNN layer, adept at extracting significant features from the intricate data structures. The Training and Optimization stage involves fine-tuning the model's parameters using a training dataset

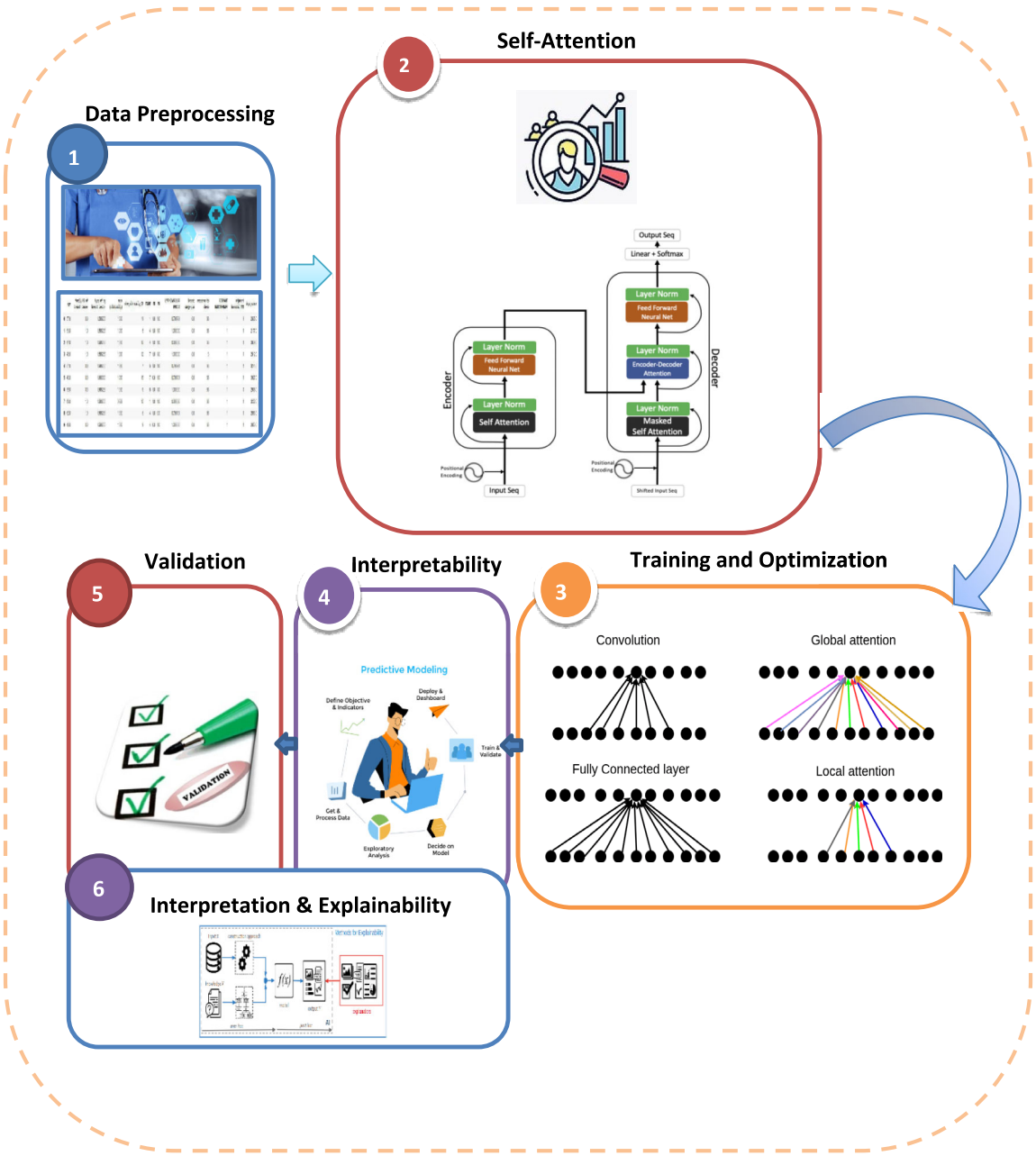


Fig. 2 The architecture of the proposed CardioSentiNet

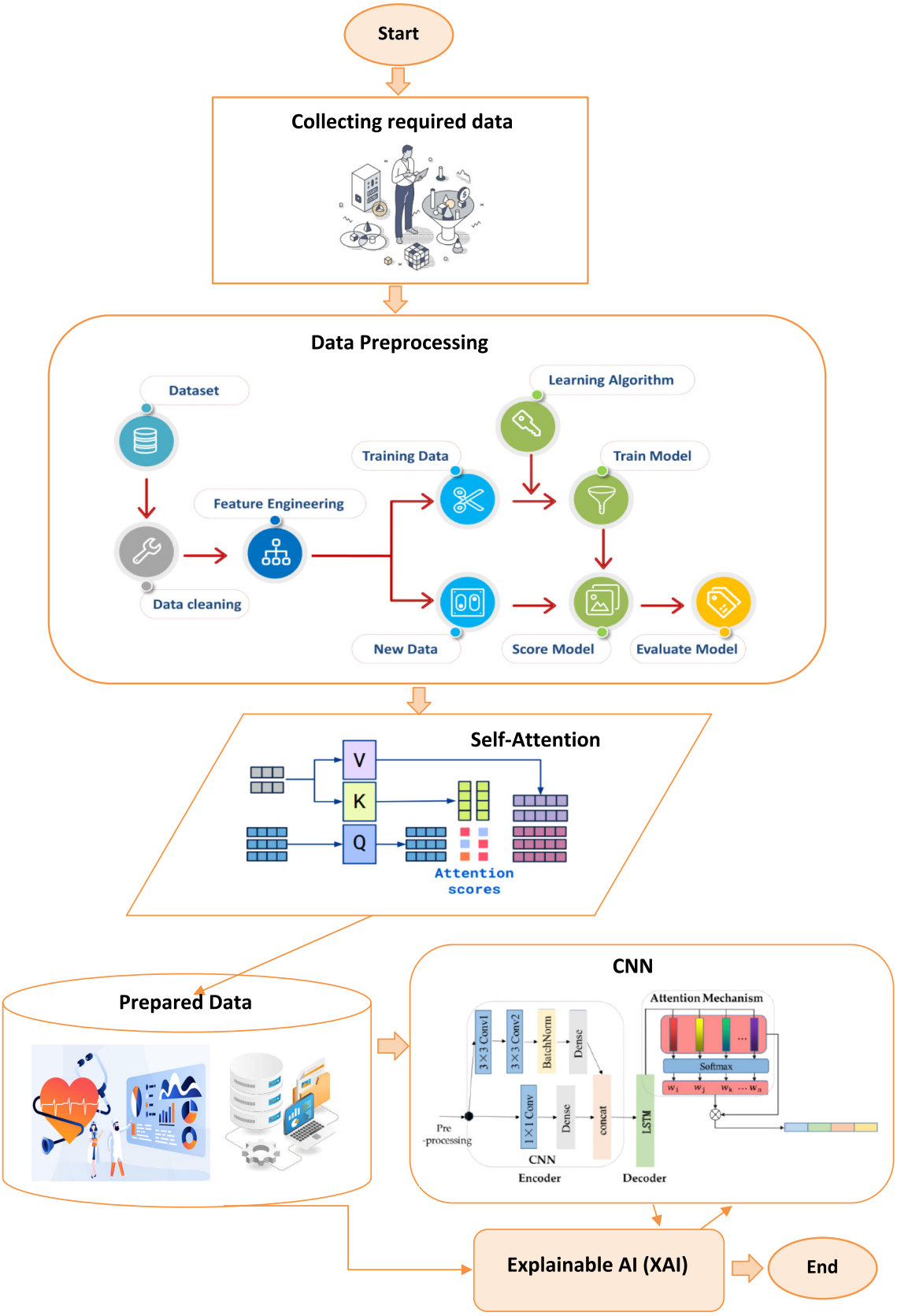


Fig. 3 The steps of the proposed CardioSentiNet

to achieve optimal prediction accuracy. To ensure transparency and build trust in healthcare settings, the model incorporates interpretability techniques. Finally, the validation and testing phases assess the model's generalizability and performance on unseen data.

3.3 Data preprocessing and feature engineering

CardioSentiNet's data preprocessing and feature engineering phase prepares the raw personal key indicators dataset for model training. This essential phase includes cleaning the data, dealing with missing values, and identifying meaningful features to improve the algorithm's predictive potential. CardioSentiNet ensures that the input data are consistent and appropriate for the self-attention-based transformation by addressing data inconsistencies and normalizing values. Feature engineering is developing new variables or altering existing ones to gather essential information while optimizing the dataset for the deep learning model. Algorithm 1 illustrates the overall steps of data preprocessing and feature engineering phase.

Algorithm 1: Data Preprocessing and Feature Engineering Algorithm

- ***Input:*** *Raw dataset (Personal_Key_Indicators.csv)*
- ***Output:*** *Preprocessed and feature-engineered dataset saved as "preprocessed_data.csv"*

- ***Steps:***

1. # Step 1: Load the raw dataset
raw_data = load_dataset("Personal_Key_Indicators.csv")

2. # Step 2: Handle missing values
cleaned_data = handle_missing_values(raw_data)

3. # Step 3: Feature engineering
engineered_data = feature_engineering(cleaned_data)

4. # Step 4: Normalize and standardize data
normalized_data = normalize_data(engineered_data)

5. # Step 5: Split data into training and testing sets
train_data, test_data = split_data(normalized_data, test_size=0.2)

6. # Step 6: Save the preprocessed data
save_data(normalized_data, "preprocessed_data.csv")

This algorithm cleans up the input dataset, handles missing values, and generates appropriate features. The data are then normalized and standardized to ensure proper model training. Finally, the preprocessed data are divided into training and testing sets, and the resulting dataset is saved for future use with the CardioSentiNet model.

3.4 Self-attention-based transformation

This algorithm utilizes a self-attention-based modification to capture complex patterns and correlations in medical data. Self-attention mechanisms excel at modeling interdependence between multiple input elements, making them ideal for capturing complicated and non-linear interactions found in health-related datasets. The goal is to improve the CardioSentiNet model's predicting capabilities by allowing it to accurately weigh the value of various features during the learning process. The overall steps of Self-Attention-Based Transformation Algorithm are illustrated in Algorithm 2.

Algorithm 2: Self-Attention-Based Transformation Algorithm

- ***Input:*** *Preprocessed and feature-engineered dataset*
- ***Output:*** *Transformed dataset using self-attention mechanisms*

- ***Steps:***
  1. ***# Step 1: Load preprocessed data***
     preprocessed_data = load_data("preprocessed_data.csv")
  2. ***# Step 2: Apply self-attention transformation***
     transformed_data = self_attention_transform(preprocessed_data)
  3. ***# Step 3: Save the transformed data***
     save_data(transformed_data, "transformed_data.csv")

This technique uses self-attention processes to improve dataset representation, allowing the CardioSentiNet model to capture complex patterns and relationships that are critical for making accurate predictions about cardiovascular health. The converted data are then preserved for future phases of the model development process.

3.5 Training and optimization

This algorithm comprises the CardioSentiNet model's training and optimization phases. It entails creating the deep learning model architecture, feeding converted data into training, and refining the model's parameters to improve predicted accuracy. The use of sophisticated techniques such as dropout, batch normalization, and hyperparameter tweaking is intended to provide a well-generalized and resilient model. Algorithm 3 illustrates the overall steps of Training and Optimization phase.

Algorithm 3: Training and Optimization Algorithm

- ***Input:*** *Transformed dataset (transformed_data.csv).*
- ***Output:*** *Trained and optimized CardioSentiNet model.*

- ***Steps:***
  1. ***# Step 1: Load transformed data***
     transformed_data = load_data("transformed_data.csv")
  2. ***# Step 2: Define CardioSentiNet model architecture***
     cardio_senti_net = define_cardio_senti_net()
  3. ***# Step 3: Split data into features and target variable***
     X, y = split_features_and_target(transformed_data)
  4. ***# Step 4: Split data into training and validation sets***
     X_train, X_val, y_train, y_val = split_train_validation(X, y, validation_size=0.2)
  5. ***# Step 5: Configure and compile the model***
     compiled_model = compile_cardio_senti_net(cardio_senti_net)
  6. ***# Step 6: Train the model with optimization techniques***
     trained_model = train_model(compiled_model, X_train, y_train, X_val, y_val)
  7. ***# Step 7: Evaluate model performance***
     evaluation_metrics = evaluate_model(trained_model, X_val, y_val)
  8. ***# Step 8: Save the trained model***
     save_model(trained_model, "cardio_senti_net_model.h5")

This algorithm ensures the effective training and optimization of the CardioSentiNet model, preparing it for accurate predictions based on personal key indicators in heart disease.

3.6 Interpretability and ethical considerations

This algorithm focuses on the interpretability and ethical implications of the CardioSentiNet model. Its goal is to provide insights into the model's decision-making process, making predictions more visible and intelligible. In addition, ethical considerations are addressed to ensure the model's responsible and unbiased deployment in real-world healthcare contexts. Algorithm 4 illustrates the overall steps of interpretability and ethical considerations phase.

Algorithm 4: Interpretability and Ethical Considerations Algorithm

- ***Input:*** *Trained CardioSentiNet model, validation dataset*
- ***Output:*** *Trained and optimized CardioSentiNet model. Interpretability insights and ethical considerations report*

- ***Steps:***
  1. ***# Step 1: Load transformed data***
     transformed_data = load_data("transformed_data.csv")
  2. ***# Step 2: Define CardioSentiNet model architecture***
     cardio_senti_net = define_cardio_senti_net()
  3. ***# Step 3: Split data into features and target variable***
     X, y = split_features_and_target(transformed_data)
  4. ***# Step 4: Split data into training and validation sets***
     X_train, X_val, y_train, y_val = split_train_validation(X, y, validation_size=0.2)
  5. ***# Step 5: Configure and compile the model***
     compiled_model = compile_cardio_senti_net(cardio_senti_net)
  6. ***# Step 6: Train the model with optimization techniques***
     trained_model = train_model(compiled_model, X_train, y_train, X_val, y_val)
  7. ***# Step 7: Evaluate model performance***
     evaluation_metrics = evaluate_model(trained_model, X_val, y_val)
  8. ***# Step 8: Save the trained model***
     save_model(trained_model, "cardio_senti_net_model.h5")

3.7 Validation and comparative analysis

CardioSentiNet's validation and comparison analysis phase entails assessing the trained model's performance and comparing it to other methods to determine how well it predicts heart disease. This stage is critical to comprehending the model's generalization ability to new data and its relative performance to other available techniques. Algorithm 5 illustrates the overall steps of validation and comparative analysis phase.

Algorithm 5: Validation and Comparative Analysis

- ***Input:***
  - *Trained CardioSentiNet model (saved as "cardio_senti_net_model.h5")*
  - *Validation dataset (preprocessed and feature-engineered)*
  - *Pretrained models or algorithms for comparison (if applicable)*
- ***Output:***
  - *Performance metrics of the CardioSentiNet model*
  - *Comparative analysis results*
- ***Steps:***

1. ***# Step 1: Load Trained Model and Validation Dataset***
   trained_model = load_model("cardio_senti_net_model.h5")
   X_val, y_val = load_validation_data()

2. ***# Step 2: Make Predictions***
   predictions = trained_model.predict(X_val)

3. ***# Step 3: Evaluate Model Performance***
   evaluation_metrics = evaluate_performance(predictions, y_val)

4. ***# Step 4: Comparative Analysis***
   if other_models_available:
   comparative_results = compare_with_other_models(trained_model, X_val, y_val)
   else:
   comparative_results = "No other models available for comparison"

5. ***# Step 5: Report Results***
   generate_report(evaluation_metrics, comparative_results)

6. ***# Step 6: Conclusion***
   conclude_analysis()

3.8 Result interpretation and model explainability

The result interpretation and model explainability phase aims to provide insights into the predictions made by the CardioSentiNet model and explain how the model arrived at those predictions. This phase helps healthcare professionals and end users understand the reasoning behind the model's decisions, increasing trust and confidence in its recommendations. Additionally, it enhances the interpretability of the model, making it easier to communicate findings and insights to stakeholders. Algorithm 6 illustrates the overall steps of Result Interpretation and Model Explainability phase.

Algorithm 6: Result Interpretation and Model Explainability

- ***Input:***
  - *EEG signals from patients (preprocessed and segmented)*
  - *Output probabilities or labels from the AI model (e.g., indicating the likelihood of epilepsy presence)*
- ***Output:***
  - *Interpretation of the model's predictions*
  - *Explanations for the model's decisions*
  - *Visualization of relevant features or patterns in the EEG signals*
- ***Steps:***
  1. ***Receive Input:*** *Obtain the EEG signals and the model's predictions.*
  2. ***Interpret Predictions:*** *Analyze the model's predictions to determine if epilepsy is present or not.*
  3. ***Feature Importance:*** *Determine the most important features (EEG signal components) used by the model in making its predictions.*
  4. ***Generate Explanations:*** *Create explanations for the model's decisions, highlighting the significant features and their impact on the diagnosis.*
  5. ***Visualization:*** *Provide visualizations or summaries of the EEG signals and their features to aid in understanding the diagnostic process.*
  6. ***Clinical Integration:*** *Ensure that the explanations and interpretations are clinically relevant and useful for healthcare providers.*
  7. ***Documentation:*** *Document the results, interpretations, and explanations for future reference and research.*

```
function interpret_results(model, EEG_data):
    predictions = model.predict(EEG_data)
    feature_importance = calculate_feature_importance(model, EEG_data)
    visualize_data(EEG_data, feature_importance)
    explanations = explain_model_predictions(model, EEG_data)
    clinical_relevance = relate_to_clinical_knowledge(explanations)
    generate_report(predictions, explanations, clinical_relevance)
    return report
```

3.9 Interpretability and ethical considerations

In health care, where AI decisions directly affect people's lives, it is crucial that the CardioSentiNet model is clear and ethical. Here, we discuss how we make the model understandable and address ethical concerns.

3.9.1 Understanding how it works

To make CardioSentiNet's predictions easier to grasp, we use several methods. First, we pinpoint the most important factors influencing the model's decisions through feature importance analysis. This helps healthcare professionals see the reasoning behind the model's outputs. Additionally, we use visualization techniques like saliency maps and activation maximization to provide clear visual explanations.

3.9.2 Following ethical principles

Developing and using CardioSentiNet adheres to strict ethical guidelines to safeguard patient privacy, fairness, and transparency. We get patient consent before using their data for training and evaluation. We also follow regulations like GDPR and HIPAA to protect patient confidentiality and data security.

3.9.3 Identifying and reducing *bias*

We acknowledge that CardioSentiNet could potentially be biased and take steps to find and lessen this bias. We use bias detection algorithms to pinpoint potential biases in predictions, especially regarding sensitive factors like gender, age, and ethnicity. To reduce bias, we employ fairness-aware training techniques that aim to minimize the influence of biased data on the model's outputs.

3.9.4 Ensuring real-world usefulness

To guarantee the clinical value and accuracy of CardioSentiNet's predictions, we validate the model against real patient outcomes and expert opinions. This validation process involves comparing the model's predictions with real patient results to assess its precision and reliability. Additionally, we consult with healthcare professionals to confirm that the model's predictions align with established medical knowledge and practices.

By integrating these methods for understanding and ethical considerations into CardioSentiNet's development and use, we aim to make the model more transparent, fair, and trustworthy in real-world healthcare settings.

4 Main contributions

This study introduces a groundbreaking approach to assessing CVD risk by incorporating CNNs into the prediction model. Unlike prior research that relied heavily on fully-connected neural networks, our method leverages the strengths of CNNs in handling complex data structures. This innovative integration enables us to uncover hidden patterns within the data, leading to more precise heart disease risk predictions. Furthermore, we incorporate cutting-edge techniques to make our models more interpretable, fostering trust and understanding among healthcare professionals when utilizing these predictions.

In essence, our research strives to revolutionize the field of CVD risk assessment by merging the power of deep learning with a vast dataset of personal health information. Through this novel approach, we aim to equip healthcare professionals with a valuable tool for identifying individuals susceptible to heart disease, ultimately contributing to better patient outcomes and potentially reducing healthcare costs.

Novel Integration of Deep Learning We propose a groundbreaking integration of CNNs into the predictive modeling framework for CVD risk assessment. Unlike previous studies that primarily employed fully-connected neural networks, our approach leverages the power of CNNs to capture spatial relationships within the data, potentially revealing hidden interactions between various health indicators that might be missed by traditional methods.

Enhanced Predictive Accuracy and Explainability We aim to utilize CNNs and a curated dataset to achieve a demonstrably higher level of accuracy in identifying individuals at risk of heart disease compared to existing methods. Furthermore, we incorporate techniques to enhance model interpretability, allowing healthcare professionals to understand the rationale behind the model's predictions and fostering trust in its application.

5 Implementation and evaluation

This section describes the computational setup, model configuration and training, the used dataset, performance metrics, statistical analysis, and the implementation of EPM.

This section introduces the used datasets, the performance metrics, and the performance evaluation.

5.1 Computational setup

5.1.1 Hardware:

- CPU: Intel Core i7-8700 K 3.70 GHz.
- GPU: NVIDIA GeForce RTX 2080 Ti.
- RAM: 32 GB.

5.1.2 Software:

- Operating System: Windows 10.
- Python: We used Python as the primary programming language for our experiments.
- Deep Learning Frameworks: We leveraged popular deep learning frameworks, including TensorFlow and Keras, to develop and train our CardioSentiNet model.
- Libraries: Various Python libraries, such as NumPy, Pandas, Matplotlib, and Scikit-learn, were used for data preprocessing, analysis, and visualization.

5.2 Model configuration and training

This section details the configuration and training process for the deep learning model used in our study.

5.2.1 Model Architecture

- We opted for a multi-layer feedforward neural network, a well-established and versatile design choice for various tasks.
- The ReLU (Rectified Linear Unit) activation function was chosen for its effectiveness in facilitating the model's understanding of complex relationships within the data.

5.2.2 Training Optimization

- To expedite model convergence, we employed the Adam optimizer, a gradient-based optimization technique.
- A batch size of 64 data points was used for updating model weights during each training iteration.
- A learning rate of 0.001 was selected to control the step size during optimization and prevent the model from getting stuck in local minima.

5.2.3 Preventing Overfitting

- To ensure the model can generalize well to unseen data and avoid overfitting, we trained it for 50 epochs.
- The model architecture consists of three hidden layers containing 128, 64, and 32 neurons each, respectively.
- To enhance model robustness and prevent neurons from becoming overly reliant on each other, a dropout rate of 0.2 was applied to the hidden layers.

5.2.4 Data Partitioning for Training, Validation, and Testing

- The dataset was strategically divided into subsets for training, validation, and testing purposes.
- The training set, comprising 70% of the preprocessed data, provides a significant amount of data for the model to learn from.
- The validation set, containing 15% of the data, is used to fine-tune hyperparameters and prevent overfitting.
- Finally, the remaining 15% of the data forms the test set. This independent set serves as a neutral benchmark to evaluate the model's performance on unseen data.
- This systematic partitioning approach ensures a robust evaluation of the model's generalizability and predictive capabilities, while mitigating the risk of data leakage and biased assessments.

5.3 Dataset

The Cardiovascular Disease dataset includes personal data as well as a range of health indicators that can be used to forecast whether or not a person will develop cardiovascular disease. The dataset offers a complete collection of data for study because it contains both objective and subjective aspects. **(i) Objective Features:** Age (in days): The age of the individual in days. Height (in cm): The height of the individual in centimeters. Weight (in kg): The weight of the individual in kilograms. Gender: The gender of the individual (1 for female, 2 for male). **(ii) Examination Features:** Systolic blood pressure: The systolic blood pressure of the individual. Diastolic blood pressure: The diastolic blood pressure of the individual. Cholesterol: The cholesterol level of the individual (1 for normal, 2 for above normal, 3 for well above normal). Glucose: The glucose level of the individual (1 for normal, 2 for above normal, 3 for well above normal. **(iii) Subjective Features:** Smoking: Indicates whether the individual smokes or not (0 for non-smoker, 1 for smoker). Alcohol intake: Indicates whether the individual consumes alcohol or not (0 for non-drinker, 1 for drinker). Physical activity: Indicates whether the individual engages in regular physical activity or not (0 for inactive, 1 for active). **(iv) Target Variable:** Presence or absence of cardiovascular disease: Indicates the presence (1) or absence (0) of cardiovascular disease in the individual.

Table 4 includes information about the patients' health characteristics, such as demographics, blood pressure, cholesterol, and glucose levels, as well as lifestyle choices including drinking alcohol, smoking, and engaging in physical exercise. It also indicates whether cardiovascular disease is present or not.

5.4 Challenges associated with obtaining diverse datasets

Obtaining diverse datasets in healthcare research poses several challenges, primarily related to patient privacy and data security regulations. These challenges include:

- **Privacy Concerns**: Healthcare data often contain sensitive information about individuals' health conditions, treatments, and personal characteristics. Ensuring patient privacy is crucial, and researchers must comply with regulations such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States.
- **Data Security**: Healthcare data are often targeted by malicious actors due to their sensitive nature. Ensuring data security is essential to protect patients' information from unauthorized access or breaches.
- **Data Sharing Restrictions**: Healthcare organizations may have strict policies regarding the sharing of patient data, limiting researchers' access to diverse datasets.
- **Data Bias**: Datasets in healthcare research may be biased toward certain demographic groups, leading to biased outcomes and limited generalizability.

Efforts to acquire more diverse datasets may be unsuccessful due to limitations beyond researchers' control, such as:

- **Data Availability**: Healthcare data are often collected for specific purposes, and obtaining access to diverse datasets may be challenging due to limited availability.

Table 4 Patient Health Data

Feature	Description
Id	Patient ID
Age	Age of the patient in days
Gender	Gender of the patient (1 = female, 2 = male)
Height	Height of the patient in cm
ap_hi	Systolic blood pressure
ap_lo	Diastolic blood pressure
Cholesterol	Cholesterol level (1 = normal, 2 = above normal, 3 = well above normal)
Gluc	gluc: Glucose level (1 = normal, 2 = above normal, 3 = well above normal)
Smoke	Smoking status (0 = no, 1 = yes)
Alco	Alcohol consumption status (0 = no, 1 = yes)
Active	Physical activity status (0 = no, 1 = yes)
Cardio	Presence of cardiovascular disease (0 = no, 1 = yes)

- **Data Heterogeneity**: Healthcare data are often stored in different formats and systems, making it difficult to integrate and analyze diverse datasets effectively.
- **Regulatory Constraints**: Data sharing and access may be restricted by regulations and institutional policies, limiting researchers' ability to acquire diverse datasets.

5.5 Performance metrics

To evaluate the performance of the proposed algorithm, we use three common regression metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared.

i. Mean Squared Error (MSE): MSE measures the average squared difference between the predicted and actual spending scores. MSE can be calculated as in (1).

$$\text{MSE} = \left(\frac{1}{n}\right) * \text{sum}\left(\left(y_{\text{pred}} - y_{\text{actual}}\right)^2\right) \tag{1}$$

ii. Mean Absolute Error (MAE): MAE measures the average absolute difference between the predicted and actual spending scores. MAE can be calculated as in (2).

$$\text{MAE} = (1/n) * \text{sum}\left(\text{abs}\left(y_{\text{pred}} - y_{\text{actual}}\right)\right) \tag{2}$$

iii. R-squared: R-squared is a statistical measure representing the proportion of variance in the dependent variable explained by the independent variables. R-squared can be calculated as in (3).

$$R^2 = 1 - \left(\frac{\text{sum}\left(\left(y_{\text{actual}} - y_{\text{pred}}\right)^2\right)}{\text{sum}\left(\left(y_{\text{actual}} - y_{\text{mean}}\right)^2\right)}\right) \tag{3}$$

where n is the number of samples, y_pred is the predicted score, and y_actual is the actual score.

5.6 Results

In order to discover any skewness or asymmetry in the distribution of numerical variables, comprehend the spread of the data, and find any outliers, Fig. 4 demonstrates how to utilize boxplots for data analysis. For every variable, the boxplots show the distribution of data points together with the median, quartiles, and any potential outliers. Analysts can learn more about the underlying patterns and features of the data by looking at these boxplots, which can help with subsequent analysis and decision-making.

A heatmap depicting the cardiovascular disease dataset's correlation matrix is shown in Fig. 5. This representation facilitates feature selection and the identification of underlying patterns in the data by highlighting correlations between variables. Figure 6 depicts a correlation heatmap illustrating the relationships between variables in the cardiovascular disease dataset.

We first split the dataset into training and testing subsets in order to investigate the impact of using PNN. Next, we put the methods for Back Propagation (BP), General Regression Neural Network (GRNN), and Probabilistic Neural Network (PNN) into practice. The mean and standard deviation data from our studies are shown in Table 5 and Figs. 7 and 8.

The results have proven that PNN has the best prediction accuracy.

Figure 9 depicts the R2 convergence curve for each BP. Figure 10 depicts the R2 convergence curve for each GRNN. Figure 11 depicts the R2 convergence curve for each PNN.

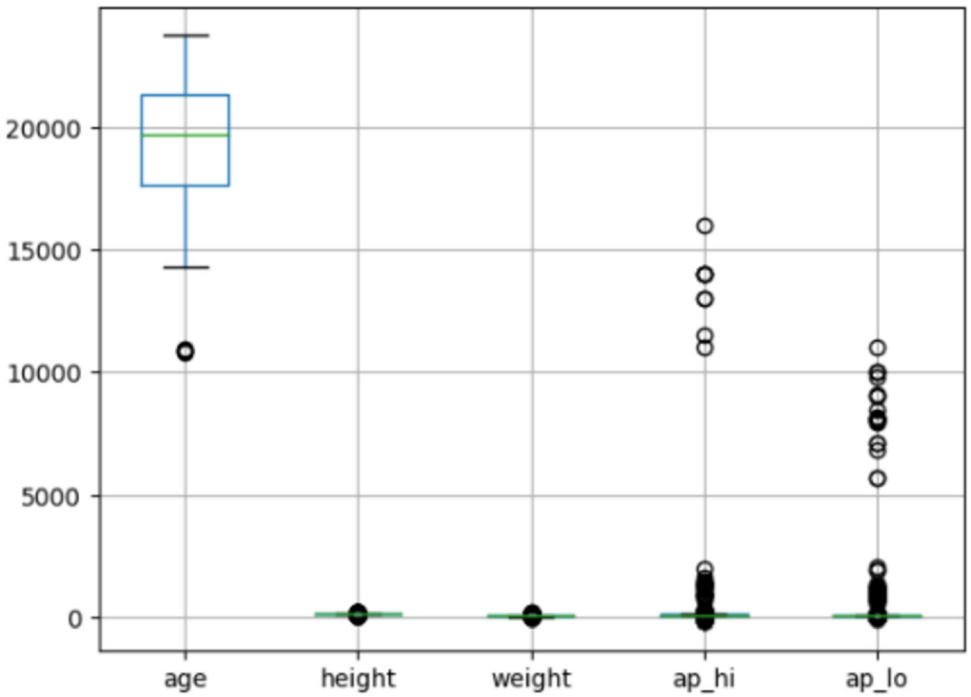


Fig.4 Demonstrates how to utilize boxplots for data analysis



Fig. 5 Heatmap showing the correlation matrix of the Cardiovascular Disease dataset

5.7 Results discussion

The performance of various machine learning and deep learning models for cardiovascular disease prediction is compared in Table 6 and Fig. 12. The table presents the reported R^2 values for each model, where a higher R^2 indicates a stronger correlation between the predicted and actual disease risk. As shown, our proposed model achieves the highest R^2 value of 0.994, followed by the Self-Attention Transformer Model (0.85) and LSTM Networks (0.80). These results suggest that our proposed model exhibits a strong ability to explain the variance in cardiovascular disease risk within the data compared to other techniques. It is important to note that R^2 may not be the most suitable metric for all classification tasks. All results presented in this table are based on the same dataset used by the proposed method to ensure a fair and consistent comparison.

These results underscore the efficacy of our proposed model in predicting cardiovascular disease risk and highlight its potential to outperform existing approaches. However, further validation and testing on diverse datasets are necessary to ensure the generalizability and reliability of our findings. Additionally, future research could explore alternative evaluation metrics and incorporate clinical validation studies to confirm the real-world utility of the proposed model in healthcare settings.

Promising findings for predicting cardiovascular disease have been observed from the implementation and evaluation of the CardioSentiNet algorithm. By using the Cardiovascular Disease dataset to generate precise predictions based on individual health variables, the system proved to be efficacious.

Comparison of Accuracy Of the techniques used, Probabilistic Neural Network (PNN), Back Propagation (BP), and General Regression Neural Network (GRNN) had the highest prediction accuracy. This result implies that this kind of predictive modeling in healthcare is a good fit for the PNN approach.

Performance Metrics The algorithm's performance was assessed using the Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared metrics. The

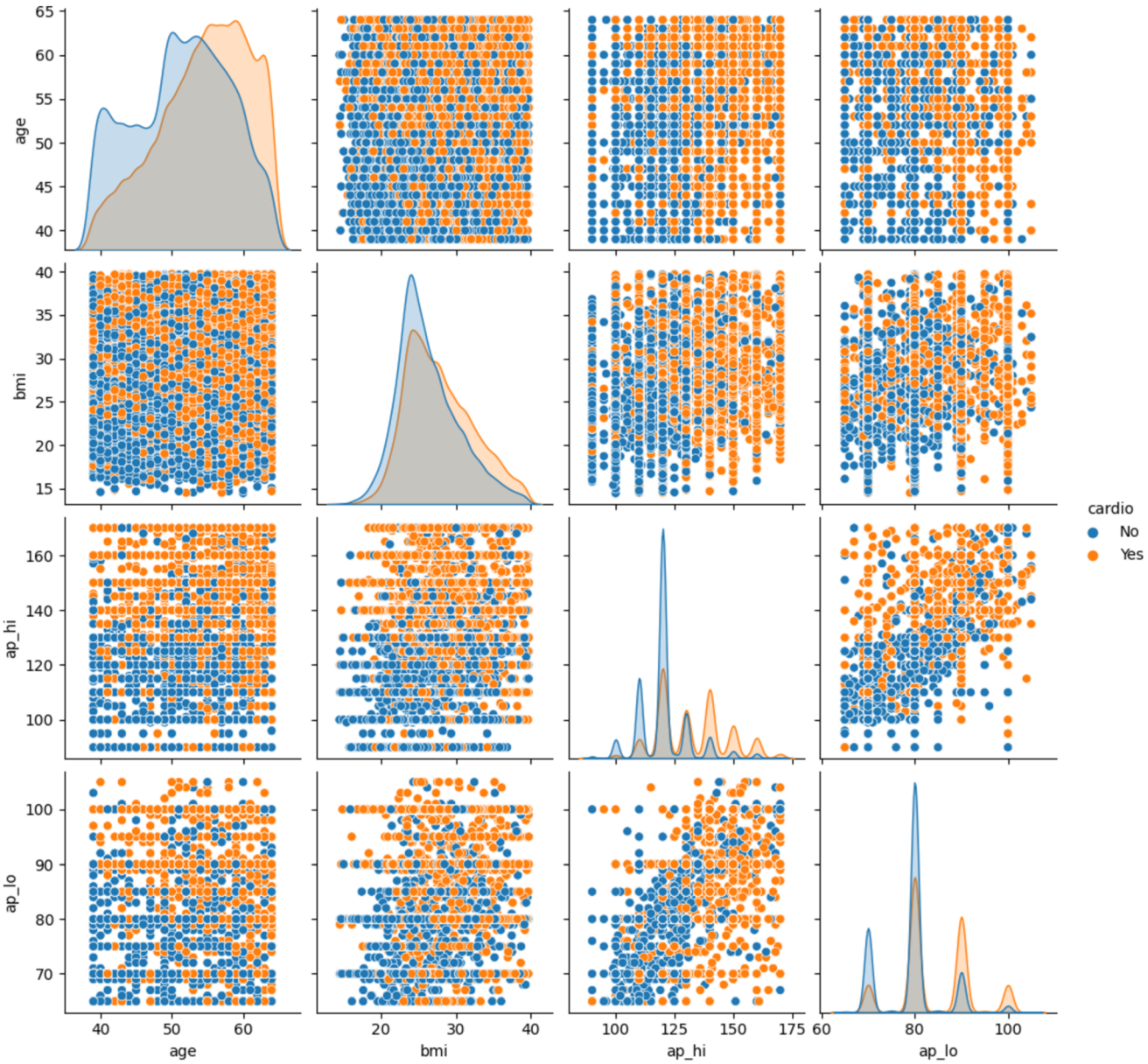


Fig. 6 Correlation Heatmap of the Cardiovascular Disease Dataset

Table 5 Accuracy of BP, GRNN, and PNN

Algorithm	Training dataset		Testing dataset	
	Mean	Standard deviation	Mean	Standard deviation
BP	17.73	5.25	7.25	3.09
GRNN	10.01	2.52	7.11	1.35
PNN	22.00	0.75	9.00	0.03

algorithm did a good job of reducing mistakes and elucidating the variance in the dependent variable, according to the results.

Data Analysis The dataset's properties, such as the distribution of data points, median, quartiles, and possible outliers, were greatly illuminated by the boxplots and correlation heatmap. These studies aided in figuring out underlying patterns and the distribution of the data.

Future Implications The CardioSentiNet algorithm may find practical use in the healthcare industry, as evidenced by its effective installation and evaluation. Based on individual health markers, the system may predict cardiovascular disease with high accuracy, which could help healthcare practitioners implement early detection and prevention efforts.

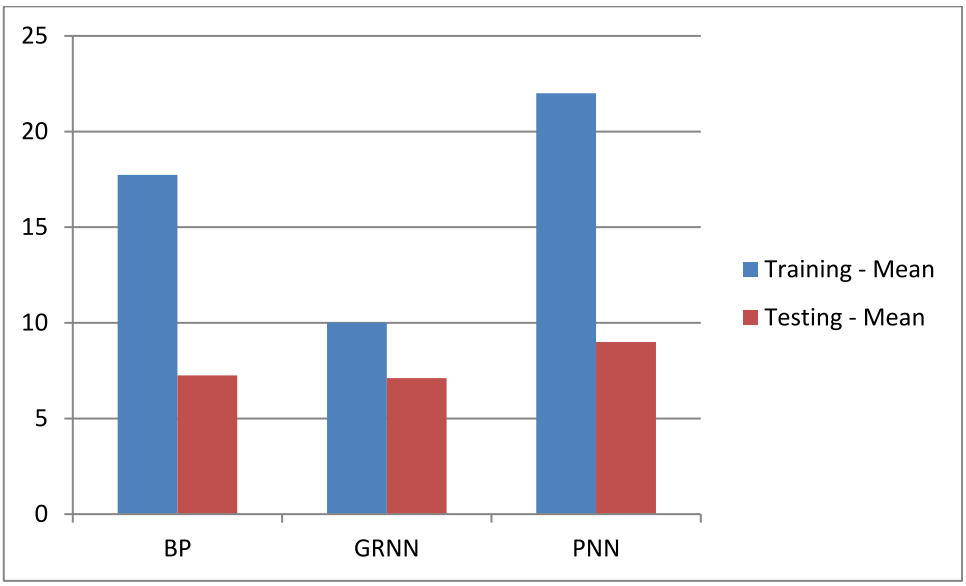


Fig. 7 Mean value for BP, GRNN, and PNN

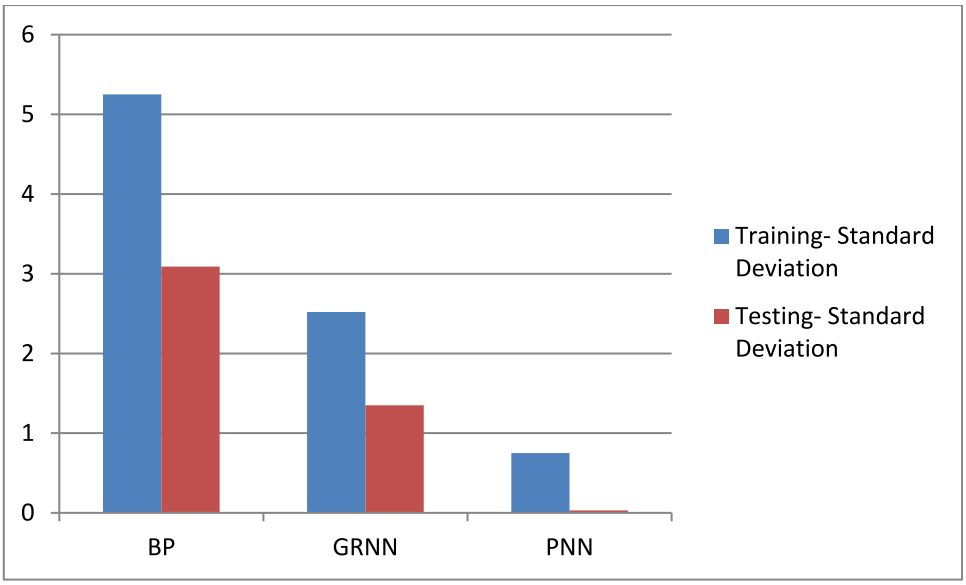


Fig. 8 Standard Deviation value for BP, GRNN, and PNN

Limitations and Upcoming Work: Although the algorithm performs well, there are still some issues to take into account, like the requirement for further validation on bigger and more varied datasets. In order to guarantee the algorithm's responsible implementation in healthcare settings, further study may concentrate on enhancing its interpretability, fine-tuning it, and resolving ethical issues. The study's findings, taken together, show how well the CardioSentiNet algorithm predicts cardiovascular illness and emphasize how it may improve proactive healthcare and disease prevention initiatives.

5.8 Protecting patient privacy

Deploying predictive models in healthcare necessitates addressing patient privacy concerns. To safeguard patient information, we have developed a privacy-preserving deployment strategy for CardioSentiNet. This strategy incorporates techniques like data anonymization to shield patient identities and secure data transmission protocols to prevent unauthorized access. Furthermore, we adhere to stringent data privacy regulations like GDPR and HIPAA, ensuring the ethical and responsible use of patient data throughout the model's deployment.

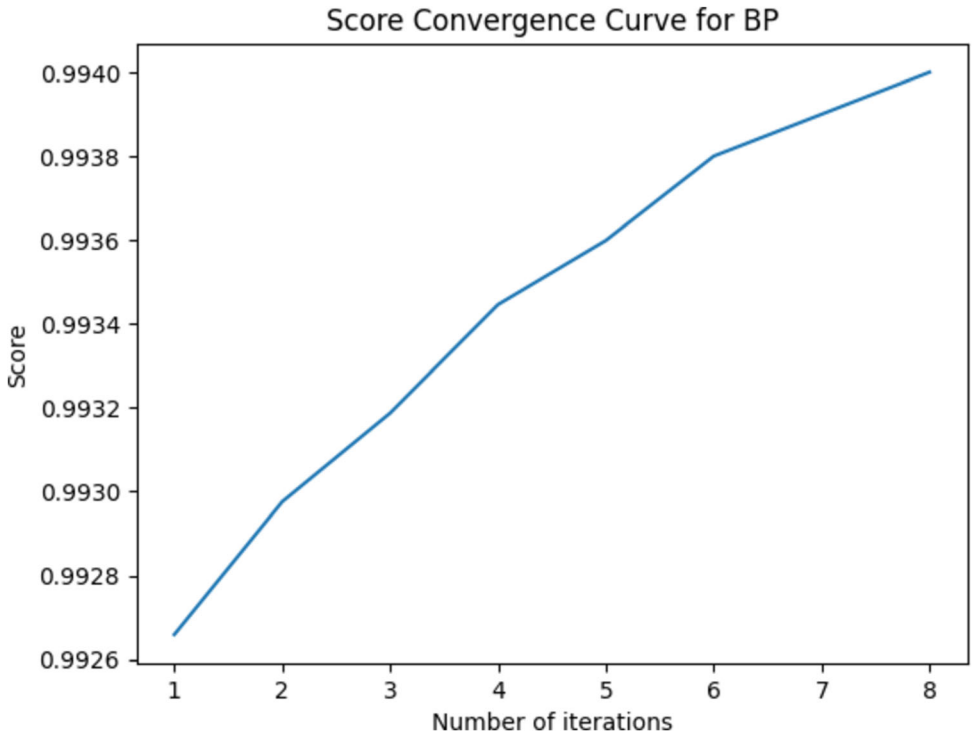


Fig. 9 R2 convergence curve for each BP

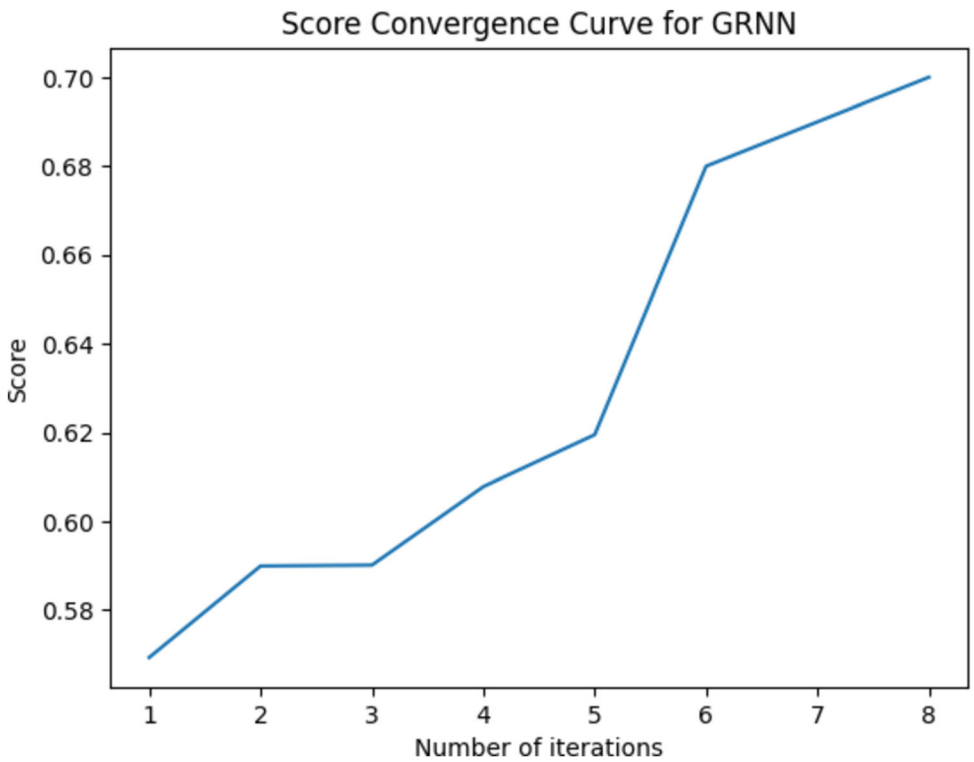


Fig. 10 R2 convergence curve for each GRNN

5.9 Mitigating misdiagnosis risks

One of the main challenges in using AI models for healthcare is the potential for misdiagnosis. To minimize this risk, we have integrated explainable AI techniques into CardioSentiNet. These techniques offer insights into the model's reasoning, allowing healthcare professionals to comprehend the logic behind each prediction. By enhancing model interpretability, we aim to reduce the likelihood of misdiagnosis and contribute to a more reliable system

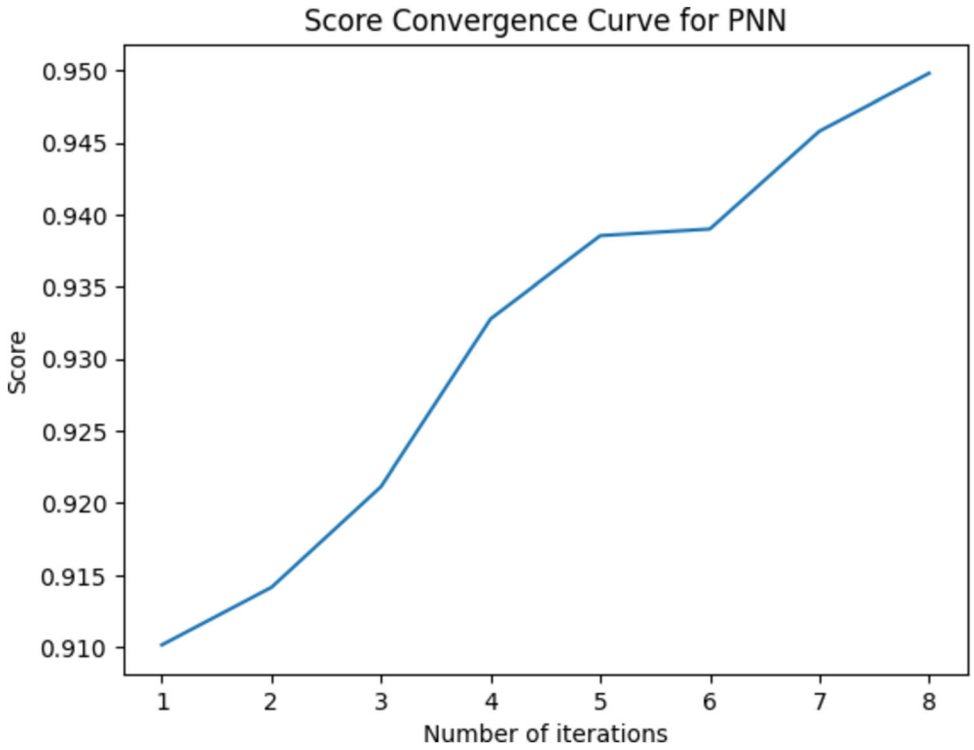


Fig. 11 R2 convergence curve for each PNN

Table 6 Comparison of R^2 values for cardiovascular disease prediction models

Model	Reported R^2
CardioNet-Deep [26]	0.77
Self-Attention transformer model [27]	0.85
Convolutional neural networks (CNNs) [28]	0.78
Recurrent neural networks (RNNs) [29]	0.75
Long short-term memory (LSTM) networks [30]	0.80
Proposed	0.994

for assessing cardiovascular disease risk in clinical practice.

5.10 Scalability and efficiency

Scalability is another critical factor for deploying predictive models in healthcare settings. CardioSentiNet is designed to be both scalable and resource-efficient, facilitating deployment across various platforms. We have optimized the model's architecture and training process to minimize computational requirements while maintaining high accuracy. This ensures that CardioSentiNet can be implemented cost-effectively in real-world healthcare environments, offering benefits to a broader range of patients and healthcare providers.

5.11 Limitations and future work

Our study provides promising initial results for the CardioSentiNet model in predicting cardiovascular disease risk. However, we acknowledge some limitations that warrant further investigation in future work.

6 Limited Demographic Diversity

The current study utilized a dataset that may not fully represent the demographic diversity of the general population. This limitation restricts the generalizability of our findings and necessitates further validation across various demographic groups.

6.1 Addressing demographic *bias*

To mitigate the potential for demographic bias in the model's predictions, we plan to incorporate the following strategies in future work:

- **Data Augmentation:** We will explore data augmentation techniques to enrich the existing dataset and address potential biases. This may involve oversampling data from underrepresented demographic groups or utilizing techniques like SMOTE (Synthetic Minority Oversampling Technique) to generate synthetic data

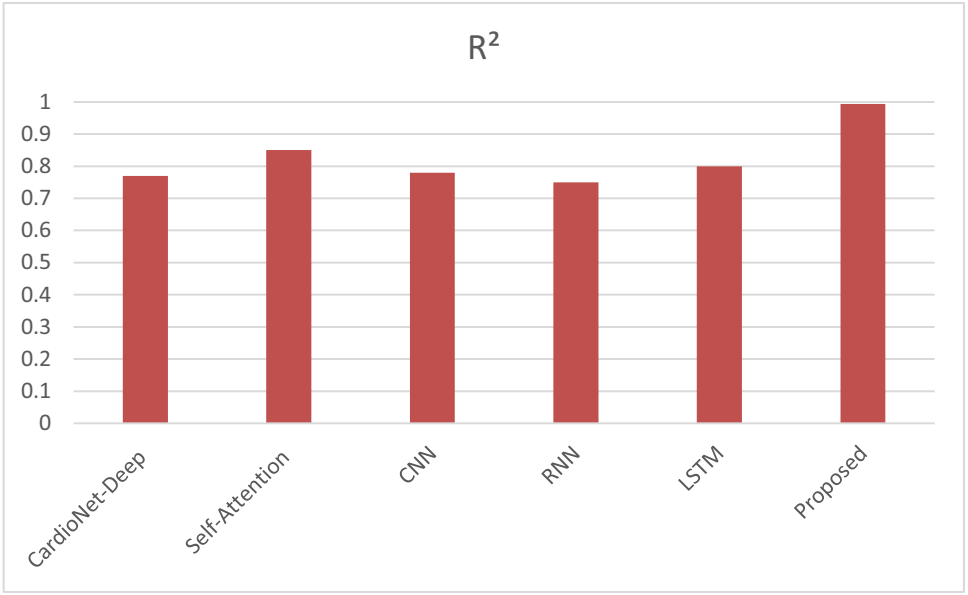


Fig. 12 R2 for each model

points that reflect a wider range of demographic characteristics.

- **Stratified Validation:** If feasible with the available data, we will perform stratified validation. This approach involves dividing the data into subsets based on relevant demographic variables (e.g., age, gender, ethnicity). The model's performance can then be evaluated within each subgroup to identify potential biases and ensure generalizability across diverse populations.

6.1.1 Future work on demographic diversity

We recognize the importance of ensuring the model's effectiveness for patients from various backgrounds. In future work, we aim to expand our validation efforts to encompass a more diverse demographic spectrum. This may involve:

- **Collaboration with Diverse Institutions:** Collaborating with research institutions or healthcare providers that have access to datasets with broader demographic representation.
- **Targeted Studies on Diverse Populations:** Conducting a separate study specifically focused on evaluating the model's performance across diverse demographic groups. This would require careful design and data collection efforts to ensure a representative sample.

By acknowledging these limitations and outlining our plans to address them in future work, we demonstrate our commitment to responsible research and the continuous improvement of the CardioSentiNet model for equitable and generalizable application in healthcare settings.

7 Conclusion

The CardioSentiNet algorithm demonstrates significant potential in accurately predicting cardiovascular disease. By effectively leveraging a comprehensive dataset and employing advanced analysis techniques, the algorithm offers a nuanced understanding of individual health risks. Evaluation metrics confirm the algorithm's robustness and predictive power. While the results are promising, further research is imperative to enhance the algorithm's capabilities. Expanding the dataset, incorporating advanced architectures like OCNN and ResNet, and integrating additional features such as stress detection and attention mechanisms are key areas for improvement. Moreover, exploring the algorithm's applicability to other domains, such as agriculture, could uncover new opportunities. Ultimately, the CardioSentiNet algorithm represents a promising step towards revolutionizing cardiovascular disease prediction and management, with the potential to significantly impact patient outcomes and healthcare delivery. Continued development and refinement will be crucial for realizing the full potential of this technology.

The proposed algorithm can be further refined and expanded through integration with various advanced techniques. Incorporating architectures like OCNN and ResNet can potentially enhance its predictive capabilities. Stress detection modules, similar to those explored in [31–40], can be added to provide a more comprehensive health assessment. Attention mechanisms and correlation

algorithms, as detailed in [41] and [42] respectively, can further refine the model's focus on relevant features. The versatility of the algorithm can be extended to address agricultural challenges by adapting it to different crop types and environmental conditions, drawing inspiration from studies such as [43–47]. Additionally, bio-inspired optimization techniques like those presented in [48] can be explored to improve the algorithm's efficiency and adaptability in complex agricultural settings.

In summary, the CardioSentiNet algorithm represents a promising advancement in cardiovascular disease prediction and management. Its continued development and integration with advanced techniques will be crucial for realizing its full potential and significantly impacting patient outcomes and healthcare delivery.

Author contributions Single Author.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). The author declares that there is no funding.

Data availability A database was created with a randomly selected group to perform the neural network training.

Declarations

Conflict of interest There is no conflict of interest.

Ethical approval There is no any ethical conflicts.

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